



# Factors influencing the working temperature of quantum dot light-emitting diodes

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**Abstract:** Quantum dot light-emitting diodes (QLEDs) possess huge potential in display due to their outstanding optoelectronic performance; however, severe degradation during operation blocks their practical applications. High temperature is regarded as one of major factors causing degradation. Therefore, a systematical study on the working temperature of QLEDs is very essential and urgent for the development of high stable QLEDs. In this work, different influence factors such as the electro-optic conversion efficiency (EOCE), voltage, current density, active area, substrate size, substrate type and sample contact are discussed in detail on the working temperature of QLEDs. The research results show that the working temperature of general QLEDs under normal operation conditions is usually smaller than 75 °C when the ambient temperature is 25 °C. However, temperature of QLEDs working under extreme conditions, such as high power or small substrate size, will exceed 100 °C, resulting in irreversible damage to the devices. Moreover, some effective measures to reduce the working temperature are also proposed. The analysis and discussion of various influencing factors in this work will provide guidance for the design of stable QLEDs and help them work at a safer temperature.

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## 1. Introduction

Quantum dot light-emitting diodes (QLEDs), usually using II-VI group, III-V group, or perovskite quantum dots (QDs) as emitting materials, have emerged in display application for their extraordinary merits, including tunable spectrum, high color saturation, low energy consumption, and simple solution processability [1–9]. Benefiting from the evolution of efficient QD emitters and rational QLED architectures, the external quantum efficiency (EQE) of red QLED has achieved 30.9% [10], while green and blue QLEDs have also reached a high EQE as 22.9% [11] and 19.8% [12] respectively. QLEDs are therefore considered a promising candidate for next-generation displays [13–15]. However, the degradation has limited the long-term capability of QLEDs in terms of brightness, efficiency and lifetime as a result of the QDs are degraded by oxygen [16,17], water [17,18], high temperature [19,20], etc. Especially for the perovskite QDs, they are farther away from practical application in QLEDs, because they have worse tolerance to the influencing factors mentioned above comparing with II-VI or III-V group QDs [21–24]. To address these issues, many strategies have been developed, such as core-shell optimization

[25,26], surface passivation [27,28], ligand optimization [22,29], etc., to protect QDs from oxygen and water effectively.

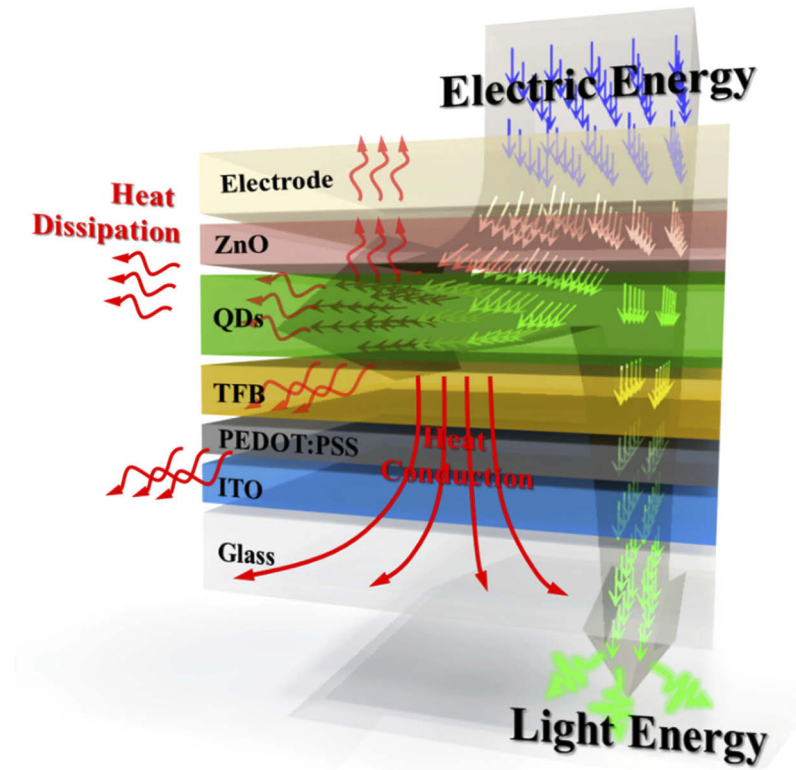
Besides of the effect of oxygen and water, high temperature also will result in severe degradation of QLEDs. There have been extensive researches on the thermal effect on QDs, especially the photoluminescence (PL) decline [30–33]. An appreciable, linear, and reversible change (1.3% per °C) of PL intensity for CdSe/ZnS QDs from 275 K up to 315 K has been reported by Walker et al. [32]. Then, Liu et al. have discussed the irreversibility of the PL quenching and shift of the CdSe/ZnS QDs. After temperature decreased to 280 K from 384 K, the PL intensity of QDs was less than 50% of the one at 280 K before heating for all samples in their study [33]. In addition, high temperature will cause more damage on perovskite QDs. Foley et al. have reported that the valence band maximum and conduction band minimum of perovskite shift down in energy by 110 meV and 77 meV as temperature increases from 28 °C to 85 °C [34]. The evolution of the emission spectrum with the changing of perovskite crystal structure will occur at 300~450 K [35]. Furthermore, the thermal decomposition begins with 323 K has been observed [36]. We can predict that higher temperature will cause more damage to QDs and QLEDs.

Although high EQE has been achieved for QLEDs, there still about 70% or more electrical energy injected into QLEDs will change into heat energy, resulting in high temperature in some situations when the devices are working. Therefore, a systematical study on the working temperature of QLEDs is very essential and urgent for the development of high stable QLEDs. However, there are few researches on the working temperature of different QLEDs, although there are several literatures having reported the temperature of their own devices [37,38]. The working temperature of different QLEDs varies greatly under different conditions thus individual results cannot represent most situations. In order to have a more comprehensive understanding of the working temperature, different factors, such as the electro-optic conversion efficiency (EOCE), voltage, current density, active area, substrate size, substrate type and sample contact, which can affect the working temperature of QLEDs, have been discussed in details. The results show that the general QLEDs under normal conditions, such as voltage < 5 V, current density < 500 mA/cm<sup>2</sup> and active area < 4 mm<sup>2</sup>, usually working at a safe temperature. However, in some extreme conditions as high power or small substrate size, the temperature of some QLEDs will exceed 100 °C, leading to a significant thermal degradation. After the discussion of these influencing factors, some effective measures to reduce the working temperature are also proposed in this work.

## 2. Thermal modeling

### 2.1. Generation and transfer of heat energy

As electroluminescence devices, QLEDs are expected to convert all the input electric energy into light energy, but in fact, 70% to 80% or even more energy is lost in the form of heat energy when QLEDs are working. Figure 1 shows the detailed structure of a common QLED where QDs film is used as active layer, poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) is used as a hole-injection layer (HIL), poly(9,9-dioctylfluorene-co-N-(4-butylphenyl)diphenylamine) (TFB) is used as a hole-transport layer (HTL) and zinc oxide (ZnO) is used as an electron-transport layer (ETL). The aluminum (Al) on the top of device and the indium tin oxide (ITO) on the glass substrate are electrodes of the QLED. After generated from active layer, the heat energy diffuses to the whole device by heat conduction and finally dissipated from the surface of the QLED to the surrounding. When this process occurs, the temperature of the whole QLEDs devices will rise first and finally maintain a constant temperature when reaching the thermal balance. Since it is rather difficult to achieve controlled adjustment of parameters of hundreds QLEDs by experiments, it is preferred to examine the maximum working temperature of QLEDs by verified finite element thermal simulations.



**Fig. 1.** The distribution and transformation of energy in a working QLED device.

In these simulations, the heat power is calculated by the difference of input electrical power and output optical power of QLEDs [39,40]. The input electrical power can be obtained through voltage, current density and active area. For the light generated by active layer, it is measured by an integrating sphere system at the initial simulation for reference and calculated by the input electrical power and EOCE in following simulations. The power of each part is calculated as follows [41]:

$$P_H = P_E - P_O \quad (1)$$

$$P_E = U \times J \times A \quad (2)$$

$$P_O = P_E \times \eta_{EO} \quad (3)$$

where  $P_E$  is the input electrical power of QLEDs,  $P_O$  is the optical power and  $P_H$  is the heating power.  $U$ ,  $J$  and  $A$  are the working voltage, current density and active area respectively. In this work, the EOCE  $\eta_{EO}$  is considered as the ratio of the optical power to the input electrical power which directly reflects the luminous capacity of a QLED device. In addition to electrical and optical power calculations, EOCE  $\eta_{EO}$  can also be derived from EQE  $\eta_{EQE}$ :

$$\eta_{EO} = \frac{np \, h\nu}{UI} = \eta_{inj} \, h\nu \frac{\eta_{EQE}}{eU} \quad (4)$$

In this equation,  $h$  is the Planck constant as  $6.626 \times 10^{-34}$  J·s and EQE  $\eta_{EQE}$  is defined as:

$$\eta_{EQE} = \frac{np}{ne} = \frac{e \, np}{\eta_{inj} I} \quad (5)$$

where  $np$  is the number of photons emitted from the QLEDs,  $ne$  is the number of electrons injected into the active layer per unit time which is equal to  $\eta_{inj} \frac{I}{e}$ . Here the  $\eta_{inj}$  is injection efficiency of the electrons and  $e$  is the charge of a single electron as  $1.602 \times 10^{-19}$  C.

Through the observation of the Eq. (4), EOCE of an idea QLED can be derived. For an example, a QLED with electroluminescence spectrum peak at 628 nm as we used in the experiment has good EQE of 30%. Its EOCE will be 29.68%, 11.87% and 5.93% at 2 V, 5 V and 10 V respectively. And if the EQE decreases to 20%, these values will become 19.78%, 7.91% and 3.95%. The rapid decrease of EOCE with the rise of voltage can be reflected in the Eq. (4).

Because of the temperature difference, the heat generated by QDs is transferred in the device. Both in the experiment and simulation, QLEDs have a cuboid geometry with strong symmetry. The heat generated by active layer spreads in different directions with same heat transfer mechanism. Thus, temperature of any point in the device can be derived from one-dimensional Fourier equation and the tedious calculation in three-dimensional can be handed over to simulation software of COMSOL Multiphysics. The steady-state heat transfer in the devices can be expressed by the Fourier equation as [42]:

$$q = -kS \frac{dT}{dx} \quad (6)$$

In the Eq. (6), the heat flux  $q$  is the integral of  $P_H$  to time,  $k$  is the thermal conductivity of the layer materials in QLEDs and  $S$  is the cross-sectional area through which the heat passes.  $x$  is the length along the coordinate in corresponding dimensions and limited by device boundary.  $dT/dx$  is the temperature gradient. In this case, the heat flux  $q$  is generated by a negative temperature gradient, so a negative sign should be added to the Eq. (6) to ensure that  $q$  is positive. The temperature difference is caused by the steady-state diffusion of heat, so it is related to the thermal conductivity  $k$ , cross-sectional area  $S$  and distance  $L$  between two points with the temperature difference in QLEDs. The expression is as follows:

$$(T_1 - T_2) = q \frac{L}{kS} \quad (7)$$

Since the heat is generated from the inside of QLEDs, which can be considered as an internal heat source. For a QLED with fixed temperature  $T_0$  at the edge or bottom, its temperature is equal to:

$$T = T_0 + q_g \frac{L^2}{2k} \left[ \frac{x}{L} - \left( \frac{x}{L} \right)^2 \right] \quad (8)$$

When the generated heat  $q_g$  of unit volume and unit time is fixed, the peak temperature occurs in the center of the device and is equal to:

$$T_{\max} = T_0 + q_g \cdot \frac{L^2}{8k} \quad (9)$$

Because  $q_g$  is the heat generated of unit volume,  $q_g = q/Lw\delta$ , the temperature difference from the center to the edge can be expressed as:

$$T_{\max} - T_0 = q \frac{L^2}{8kLw\delta} = q \frac{L}{8kS} \quad (10)$$

where cross-sectional area  $S$  is the product of width  $w$  and thickness  $\delta$ . Considering that there are many layers in QLEDs, the thermal conductivity  $k$  here is treated as a total equivalence value and can be calculated from the thermal conductivities  $k_1 \sim k_n$  of different layers of materials. As a

transition parameter, thermal resistances  $R_1 \sim R_n$  of different layers of materials are used here:

$$R_{total} = R_1 + R_2 + \dots + R_n = \frac{L_1}{Sk_1} + \frac{L_2}{Sk_2} + \dots + \frac{L_n}{Sk_n} \quad (11)$$

$$k = \frac{L}{SR_{total}} = \frac{L_1 + L_2 + \dots + L_n}{\frac{L_1}{k_1} + \frac{L_2}{k_2} + \dots + \frac{L_n}{k_n}} \quad (12)$$

The distances  $L_1 \sim L_n$  are the path distance in each material with different thermal conductivity. When the heat is transferred from the active area of QLEDs to the surface, the device will have convective heat transfer with the surrounding air. The convective heat transfer from the device surface to the surrounding air can be expressed by the heat transfer coefficient  $h$ , the temperature difference between the surface and the air, and the area of dissipation  $S_d$  [43].

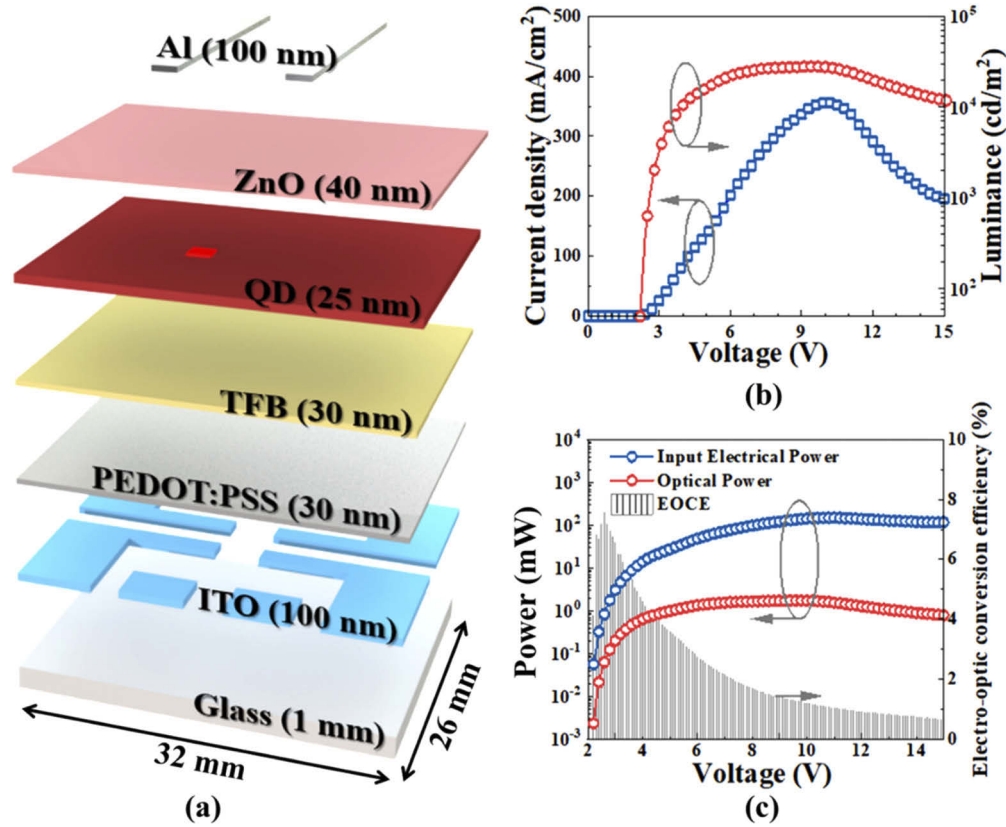
$$q = \gamma S_d (T_s - T_{air}) \quad (13)$$

The difference of heat transfer between different fluids is reflected in the value of  $\gamma$ , and  $\gamma$  can be obtained from empirical formula or theoretical relationship. In this study, since the surrounding of QLEDs is considered as the natural convection heat transfer in the laboratory,  $h$  will be set as  $8 \text{ W}/(\text{m}^2 \cdot \text{K})$  [41].

## 2.2. Thermal modeling and verification

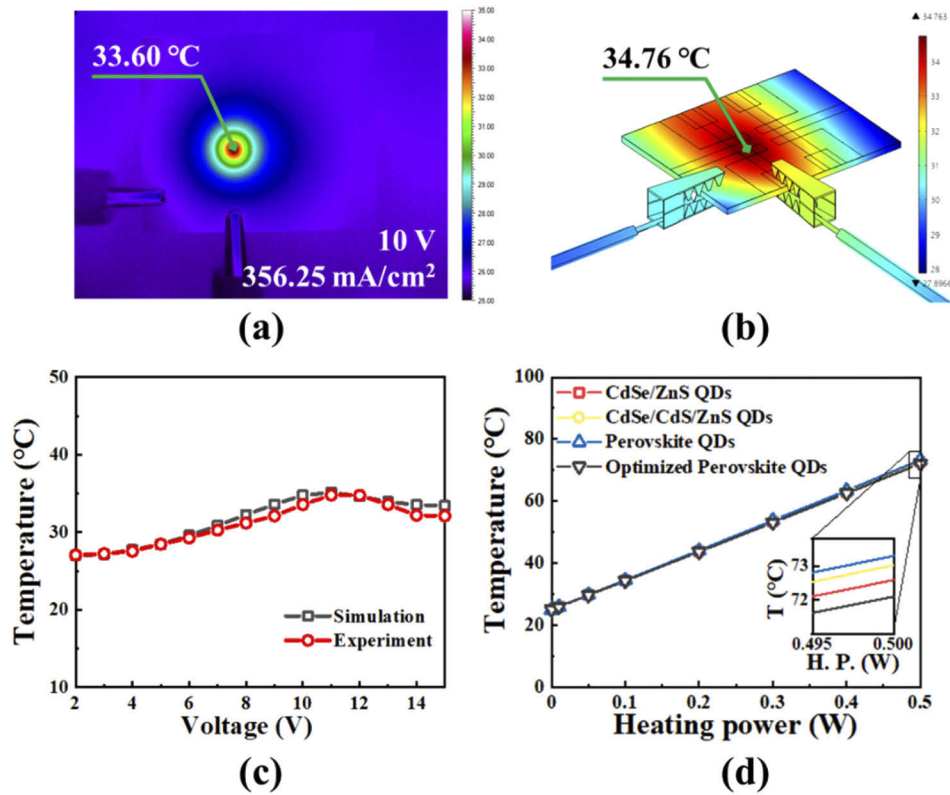
In this work, a QLED device is fabricated firstly according to the common structure and scale shown in Fig. 2(a) as a reference. The bright red area represents the active area of this QLED and is  $2 \text{ mm} \times 2 \text{ mm}$  in the reference QLED. Because it will reach different temperature when placed on different planes, the QLED is clamped by clips and suspended in the air during the experiment. It should be noted that the working temperature of QLEDs suspended in the air will be slightly higher than which contacted with planes. This phenomenon will also be discussed in detail later. During the experiment, a Keithley 2400 electrometer is used for  $J$ - $U$  characterization and a fiber integration sphere coupled with a QE-65000 spectrometer are used for light output measurements. In electricity, as shown in Fig. 2(b), this QLED exhibits an increasing current density at low voltage ( $< 10 \text{ V}$ ) whereas it shows a decreasing trend at higher voltage. This is because the device is damaged at ultra-high voltage. Meanwhile, the luminance of this QLED also has a decrease which are directly reflected in the optical power (Fig. 2(c)). EOCE keeps a decreasing trend with the increase of voltage since  $2.6 \text{ V}$ , which is mainly because it has a negative correlation with the voltage (Eq. (4)).

Figure 3(a) shows the experiment temperature field of this QLED under  $10 \text{ V}$  while the current density is  $356.25 \text{ mA}/\text{cm}^2$ . The highest temperature in the QLED is  $33.60^\circ \text{C}$ . Refer to the same structure in Fig. 2(a), a QLED model is established in the simulation. The size and thermal conductivity of each component used for thermal simulation are listed in Table 1 in the Appendix. Figure 3(b) shows the simulated temperature field of QLEDs under the same voltage and current density with the experiment. The highest temperature in the QLED is  $34.76^\circ \text{C}$ , which is very closed to the experiment data of  $33.60^\circ \text{C}$ . More comparison details of simulation and experiment can be seen in Fig. 3(c) and the relative average deviation of the temperature at each voltage point is  $1.89\%$ . It is seen that the simulated result agrees well with the reference experiment. The slightly temperature difference between simulation and experiment is mainly caused by ambient temperature and indoor air flow. When the voltage continues to rise, there is a slight drop of the working temperature of QLED. Combined with the previous analysis, this decrease is due to the reduction of the input electrical power caused by the sharp drop in the current density. In addition to the CdSe/ZnS QDs used to verify the accuracy of the model, QLEDs will also use some other active layer materials with different thermal conductivity. At present, there are few reports on the thermal conductivity of QDs, including core/shell QDs as CdSe/ZnS QDs [36],



**Fig. 2.** (a) Structure diagram of the reference QLED, (b) current density–voltage–luminance ( $J$ – $U$ – $L$ ), (c) power–voltage–EOCE ( $P$ – $U$ – $EOCE$ ).

core/shell/shell QDs as CdSe/CdS/ZnS QDs [44], perovskite QDs as CsPbBr<sub>3</sub> QDs [45] and thermal conductivity optimized perovskite QDs as MAPbI<sub>3</sub> QDs-embedded polyacrylonitrile [46]. Figure 3(d) show that the active layers with different thermal conductivities (Table 1 in the Appendix) have no obvious effect on the working temperature of QLEDs. According to Eq. (7), the thermal conductivity  $k$ , cross-sectional area  $S$  and distance  $L$  between top and bottom boundaries of the film material have influenced the temperature difference between its two adjacent layers. Because the  $L$  as the thickness of active layer material here is ultra-thin (25 nm) and the  $S$  as the device size is very large (32 mm×26 mm), it cannot cause obvious temperature change unless the thermal conductivity is also a ultra-small value. In fact, although the thermal conductivity of different materials is different, they are all in the range of 0.1 ~ 1.3 W/(m·K). This makes the influence of the thermal conductivity of active layer materials and various transport layer materials having no obvious effect on the working temperature.

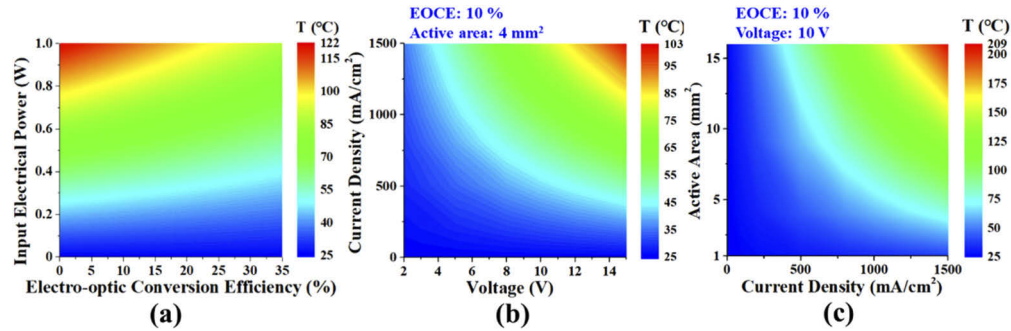


**Fig. 3.** The temperature field images of QLED worked at 356.25 mA/cm<sup>2</sup> in (a) the actual measurement and (b) the simulation. (c) The working temperature of the simulation and experiment at different voltage. (d) The influence of the active layer materials on the working temperature of QLEDs. The insert shows the detailed data when the heating power is close to 0.5 W.

### 3. Results and discussions

#### 3.1. Influence of heating power

Heating power reflects the thermal generation in QLEDs and mainly related to their input electrical power and EOCE. It can be expressed as  $P_H = P_E \times (1 - \eta_{EO})$  derived from Eq. (1) and Eq. (3). Working temperature of QLEDs under different input electrical power and EOCE has been simulated (Fig. 4(a)). The QLED model is completely refer to the structure and size shown in Fig. 2(a). 4 mm<sup>2</sup> (2 mm × 2 mm) is one of the most commonly used active area of QLEDs. The input electrical power of these QLEDs will be about 0.1 W when they are working at 5 V with the current density of 500 mA/cm<sup>2</sup>. With an EOCE of 20%, their working temperature will reach 32.50 °C. It is a safe temperature for most QLEDs. However, there are many high power QLEDs working at higher voltage and current density for their high brightness requirement. The EOCE will also decrease rapidly with the increasing voltage as discussed before. For a QLED working at 10 V with an input electrical power of 1 W, its EOCE will usually less than 5% and the working temperature of which will exceed 116.97 °C. Such a high temperature will obviously accelerate the thermal degradation of QLEDs. Higher EOCE can effectively reduce the working temperature. Once the EOCE of this QLED is increased to 30%, its working temperature will become 92.12 °C which is 24.85 °C lower than before.



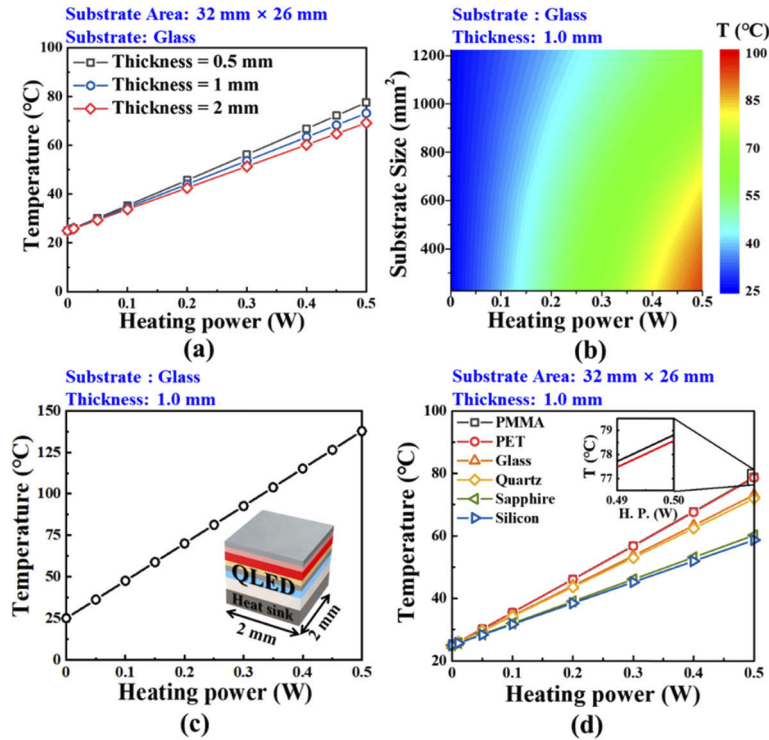
**Fig. 4.** The influence of various factors on the working temperature of QLEDs. These factors include (a) EOCe, (b) voltage and current density, (c) active area.

The working temperature can be effectively influenced by controlling the input electrical power. According to Eq. (2), the input electrical power  $P_E$  is determined by voltage  $U$ , current density  $J$  and active area  $A$ . Because of the interrelation of these three factors, active area of  $4 \text{ mm}^2$  (Fig. 4(b)) and working voltage of 10 V (Fig. 4(c)) are selected as the controlled variables in the simulation. And in these cases, the EOCe is considered as 10% which is a tradeoff value of common QLEDs. The results in Fig. 4(b) show that a QLED working under 10 V and  $1000 \text{ mA}/\text{cm}^2$  will achieve  $59.07^{\circ}\text{C}$  and most of the QLEDs have a working temperature lower than it. Indeed, a QLED will be only  $26.68^{\circ}\text{C}$  at 5 V with the current density of  $100 \text{ mA}/\text{cm}^2$ . However, if the voltage or current density continue to increase, the working temperature of QLEDs will increase with them too and some QLEDs with high voltage and current density will reach  $100^{\circ}\text{C}$ . Besides the voltage and current density, Fig. 4(c) shows that the active area can also affect the working temperature effectively. Still for the QLEDs working under 10 V and  $1000 \text{ mA}/\text{cm}^2$ , the working temperature will reach  $95.86^{\circ}\text{C}$  when the active area is adjusted from  $4 \text{ mm}^2$  to  $9 \text{ mm}^2$ . Further increasing the active area will make the working temperature reach the level that QLEDs cannot tolerate. Once the active area changes to  $16 \text{ mm}^2$ , the working temperature will exceed  $200^{\circ}\text{C}$  and before that the QLEDs will be destroyed. Combining Figs. 4(b) and (c), for QLEDs with voltage  $< 5 \text{ V}$ , current density  $< 500 \text{ mA}/\text{cm}^2$  and active area  $< 4 \text{ mm}^2$ , their working temperature will be lower than  $33.44^{\circ}\text{C}$ . This is mainly because that the input electrical power of QLEDs has kept at a low level by these three factors.

### 3.2. Influence of substrates

As discussed before, the active layers with different thermal conductivities have no obvious effect on the working temperature of QLEDs because of their thin thickness. Therefore, the substrate plays a decisive role in the heat conduction and dissipation of the whole QLED device. In this work, the influence of substrates on the working temperature of QLEDs is studied. Researchers usually use glass as the substrate material and the thickness of glass substrate is generally fixed at about 1 mm. The simulation results show that the QLEDs with thicker substrates will achieve a lower working temperature (Fig. 5(a)) because of the larger dissipation area. But the temperature difference is only  $3.53^{\circ}\text{C}$  when the QLED with the heating power of  $0.45 \text{ W}$  ( $P_E = 0.5 \text{ W}$ ,  $\eta_{EO} = 10\%$ ) changes its substrate thickness from 1 mm to 2 mm. It is difficult to have a significant difference if there is no larger change of the substrate thickness. However, if the substrate thickness is greatly increased, the optical performance of QLEDs might be affected and it is unreasonable in practical application.

Changing the substrate size will affect the working temperature of QLEDs more significantly. To verify this forecast, the relationship between different substrate size and working temperature has also been studied. As predicted, the working temperature of QLEDs rises with the decrease of



**Fig. 5.** The influence of (a) thickness, (b) substrate size and (d) the materials of substrates on the working temperature of QLEDs. And (c) shows the working temperature of QLED with 2 mm×2 mm×1 mm substrate under different heating power. The insert shows the detailed data when the heating power is close to 0.5 W.

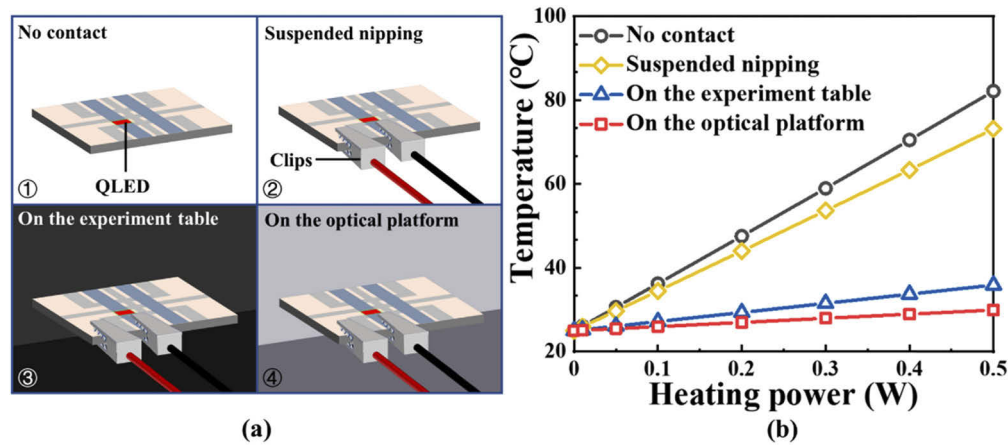
substrate size (Fig. 5(b)). For a QLED with a 32 mm×26 mm substrate, its working temperature will be 67.71 °C if its heating power is 0.45 W ( $P_E = 0.5$  W,  $\eta_{EO} = 10\%$ ). Once the substrate size changes to 225 mm<sup>2</sup> (15 mm×15 mm), its working temperature will increase to 87.56 °C. In practical application, QLEDs will use much smaller size substrate which is usually the same size to the devices. A special QLED model is built here with 2 mm×2 mm substrate (Fig. 5(c)). It has almost same device structure as what in Fig. 2(a), but it is bonding on a hypothetical heat sink rather than suspending as before. This is because the temperature of such a small QLED suspended in the air will reach 130 °C when its power is only 0.01 W. It is impractical and meaningless for it to work suspending at a higher power. Bonding on the hypothetical heat sink whose bottom is set as 25 °C, this small size QLED can also achieve a temperature of 126.51 °C when its heating power is 0.45 W. This temperature value is almost twice to the temperature of the previous model.

In addition to the size of the substrate, the material type of the substrate will also have a big influence on the working temperature. QLEDs can obtain lower working temperature by choosing a substrate with larger thermal conductivity (Fig. 5(d)). The thermal conductivities of different materials used in the simulation, such as PMMA, PET, glass, quartz, sapphire and silicon, can be found in Table 1 in the Appendix. It should be noted that the silicon substrate can only be used in top emitting QLEDs because of its opacity. PMMA and PET have the similar thermal conductivity mainly because they are both polymers, thus the working temperature of QLEDs using these two substrates cannot be distinguished easily. When the heating power is 0.45 W ( $P_E = 0.5$  W,  $\eta_{EO} = 10\%$ ), the QLEDs choosing these two type substrates will be 5.1 °C

higher than which choosing the glass substrate. Different with the polymer materials, sapphire and silicon have good heat dissipation performance. Considering the universality, the sapphire, whose thermal conductivity of is 46 W/(m·K), is a good choice to reduce the working temperature. By changing a substrate from glass to sapphire, the working temperature a QLED with heating power 0.45 W ( $P_E = 0.5$  W,  $\eta_{EO} = 10\%$ ) will be reduced 11.53 °C.

### 3.3. Influence of external contacts

The surrounding conditions also have a great influence on the working temperature of QLEDs. Considering that there is no forced air convection at the measurement platform, it usually set natural convection and 25 °C in simulation. Besides air, the surrounding conditions that affect the working temperature also contain the external contacts of QLEDs which usually include fixtures and the experimental platforms for sample placement. The most commonly used fixture in the laboratory is the clip (Fig. 6(a) ②). The results in Fig. 6(b) show that the clips influence the working temperature obviously while the QLEDs are suspended. It is mainly because the clips have a similar size with QLEDs and provide considerable area of dissipation. In addition to contact with the fixture, QLEDs will also contact with the experimental platforms sometimes. The tabletop of a common experiment table usually uses the material of epoxy resin which has an excellent chemical corrosion resistance capacity. For an optical platform, its tabletop usually uses stronger and less deformable materials, such as stainless steel. Stainless steel has a higher thermal conductivity than the epoxy resin and the details can be found in Table 1 in the Appendix. These two kinds of experimental platforms are considered here (Fig. 6(a) ③ ④). In the simulation, it is generally considered that these platforms will remain at 25 °C because their huge heat dissipation area. As can be seen in Fig. 6(b), the working temperature of QLEDs placed on the experimental platforms will be significantly lower than which of suspended situations. When the heating power  $P_H$  reaches 0.45 W ( $P_E = 0.5$  W,  $\eta_{EO} = 10\%$ ), the working temperature of the QLED suspended by the nipping of clips is 58.49 °C, but it is only 32.63 °C for the QLED on the experiment table and 28.56 °C for the QLED on the optical platform.



**Fig. 6.** (a) Diagrams of different external contacts of QLEDs and (b) the working temperature of these situations.

## 4. Conclusions

In this paper, series of comparisons are established to research the working temperature of QLEDs and its influencing factors. Among these factors, the input electrical power and substrate size are most influential. QLEDs with high power or small substrate size will easily reach a high

working temperature exceeds 100 °C which can damage these devices. EOCE of QLEDs will also have significant impact to the working temperature and a high EOCE will help the high power QLED to work at a low temperature range. In addition, keeping the voltage and current density at a low level, as well as a small active area design, will ensure the QLEDs working at a safe temperature directly. For the QLEDs with special requirements to work at a high voltage or current density, a suitable substrate size and materials also have notable heat dissipation effect. Although most of QLEDs have the working temperature smaller than 75 °C, this temperature range is still challenging for the operation lifetime of some QLEDs. To make the application of QLEDs more extensive, more attention should be paid to thermal management solutions.

## Appendix

**Table 1. Thickness and thermal conductivity used for thermal simulation<sup>a</sup>**

| Component                                              | Thickness $L$ [nm]              | Thermal Conductivity $k$ [W/(m·K)] |
|--------------------------------------------------------|---------------------------------|------------------------------------|
| Al (Aluminum)                                          | 40                              | 237                                |
| ZnO (Zinc Oxide)                                       | 40                              | 29.98                              |
| CdSe/ZnS QDs <sup>b</sup> , [36]                       | 25                              | 0.6                                |
| CdSe/CdS/ZnS QDs [44]                                  | 25                              | 0.24                               |
| CsPbBr <sub>3</sub> QDs [45]                           | 25                              | 0.115                              |
| MAPbI <sub>3</sub> QDs-embedded polyacrylonitrile [46] | 25                              | 1.25                               |
| TFB                                                    | 30                              | 0.2                                |
| PEDOT: PSS [3]                                         | 30                              | 0.2                                |
| ITO                                                    | 100                             | 1336                               |
| Glass [5]                                              | 1000000                         | 1.1                                |
| PMMA                                                   | 1000000                         | 0.18                               |
| PET                                                    | 1000000                         | 0.2                                |
| Quartz                                                 | 1000000                         | 1.46                               |
| Sapphire [5]                                           | 1000000                         | 46                                 |
| Silicon                                                | 1000000                         | 150                                |
| Component                                              | Size [mm]                       | Thermal Conductivity [W/(m·K)]     |
| Clips (Copper)                                         | About 10×4×2                    | 400                                |
| Wire (Copper)                                          | R = 0.5, L = 100                | 400                                |
| Wire coating (PVC)                                     | R = 1, L = 90                   | 0.1                                |
| Hypothetical heat sink                                 | L = W = 2, H = 1 <sup>c</sup>   | 170                                |
| Experimental table (Epoxy resin)                       | L = W = 20, H = 20 <sup>c</sup> | 0.2                                |
| Optical platform (Stainless steel)                     | L = W = 20, H = 20 <sup>c</sup> | 17                                 |

<sup>a</sup>The thermal conductivities are reference to the literatures and Wikipedia.

<sup>b</sup>The QDs material used in the experiment is CdSe/ZnS QDs (628 nm, Xingshuo Nanotech Co., Ltd.).

<sup>c</sup>The hypothetical heat sink and experimental platforms in simulation are set as board. Their size will not have a significant impact on QLEDs because platforms are set to a constant temperature 25 °C like an ideal heat sink.

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## Disclosures

The authors declare no conflicts of interest.

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